

What is a Catalanimal?

George H. Seelinger

ghseeli@umich.edu

ICERM: What is...? Seminar

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- My IPAC 2024 Lectures:
<https://www2.math.upenn.edu/~jhaglund/IPAC/2024.html>
- Anna Pun's 2021 Lectures:
<https://www2.math.upenn.edu/~jhaglund/IPAC/2021.html>

Hall-Littlewood Examples

$$\begin{aligned}H_{(4,2)} &= \text{pol} \left(\sigma \left(\frac{x_1^4 x_2^2}{1 - tx_1/x_2} \right) \right) \\&= \text{pol} \left(\sigma \left(x_1^4 x_2^2 + tx_1^5 x_2 + t^2 x_1^6 x_2^0 + t^3 x_1^7 x_2^{-1} + \cdots \right) \right) \\&= \text{pol} \left(\chi_{42} + t\chi_{51} + t^2\chi_{60} + t^3\chi_{7,-1} + \cdots \right) \\&= s_{42} + ts_{51} + t^2 s_6 .\end{aligned}$$

$$\begin{aligned}H_{(4,2,1)} &= \text{pol} \left(\sigma \left(\frac{x_1^4 x_2^2 x_3^1}{(1 - tx_1/x_2)(1 - tx_1/x_3)(1 - tx_2/x_3)} \right) \right) \\&\dots \\&= s_{421} + ts_{43} + ts_{511} + (t^2 + t)s_{52} + (t^3 + t^2)s_{61} + t^4 s_7\end{aligned}$$

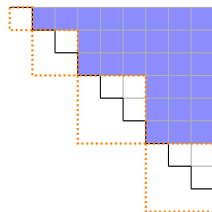
Root Ideals

	(12)	(13)	(14)	(15)
		(23)	(24)	(25)
			(34)	(35)
				(45)

	(12)	(13)	(14)	(15)
		(23)	(24)	(25)
			(34)	(35)
				(45)

Catalan Function Examples

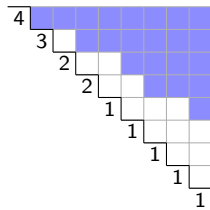
- Parabolic Hall-Littlewood polynomials $H_{\eta,\lambda} = H(R_{\eta}^+, \lambda)$



$$\eta = (1, 2, 3, 3)$$

$$R_{\eta}^+ =$$

- k -Schur functions $s_{\lambda}^{(k)} = H(\Delta^k(\lambda), \lambda)$



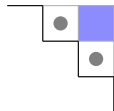
$$\Delta^4(432211111) =$$

Catalan Example

Special case: $R_+ = R_q = R_t \supseteq R_{qt}$

• $R_t \setminus R_{qt}$

■ R_{qt}



$$H(R_+, R_+, \{\alpha_{13}\}, 111)$$

$$= \text{pol } \sigma \left(\frac{x_1 x_2 x_3 (1 - q t x_1 / x_3)}{\prod_{1 \leq i < j \leq 3} (1 - q x_i / x_j) (1 - t x_i / x_j)} \right)$$

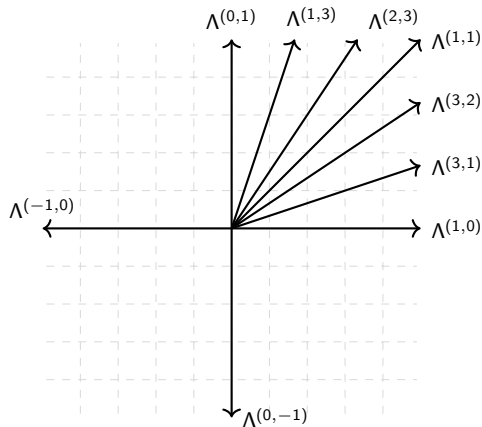
$$= \dots$$

$$= s_{111} + (q^2 + q t + t^2 + q + t) s_{21} + (q^3 + q^2 t + q t^2 + t^3 + q t) s_3$$

$$= \omega \nabla e_3$$

Elliptic Hall Algebra $\mathcal{E}$

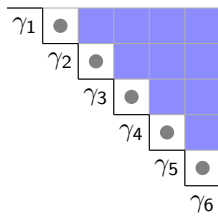
$$\Lambda^{(m,n)} = \Lambda(X^{m,n}) \cong \Lambda$$



Negut Catalanimal

Special case: $R_+ = R_q = R_t \supseteq R_{qt}$

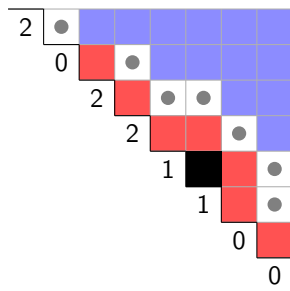
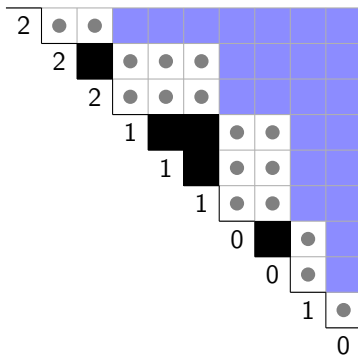
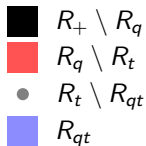
- $R_t \setminus R_{qt}$
- R_{qt}



$$H(R_+, R_+, [R_+, R_+], \gamma) = \text{pol } \sigma \left(\frac{x^\gamma \prod_{i+1 \leq j} (1 - qtx_i/x_j)}{\prod_{i < j} (1 - qx_i/x_j)(q - tx_i/x_j)} \right)$$

Some (1, 1)-Cuddly Catalananimals

Special case: $R_+ \supseteq R_q \supseteq R_t \supseteq R_{qt}$



Catalanimals and shuffle theorems

$$\begin{array}{ccc}
 \mathcal{E}^+ & \xrightarrow{\sim} & \mathcal{S} \\
 \downarrow & & \downarrow \\
 \mathcal{E}^+ \curvearrowright 1 \in \Lambda & \equiv & \begin{array}{l} \text{Tame} \quad \text{Catalanimal} \\ \text{(explicit description)} \end{array} \\
 \parallel & & \downarrow \text{Cauchy formula} \\
 \text{Algebraic quantity} & = & \text{Combinatorial quantity}
 \end{array}$$

Mystery

